A Critical Review on Video-Based Continuous Sign Language Recognition

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Abstract. Sign language bridges communication gaps, enabling hearing impaired people to understand and connect among themselves. Sign language recognition (SLR) is an automated algorithm that translates hand gestures from hearing impaired into text for facilitating interactions with normal people. In recent years with the explosion of research in technologies like computer vision, deep learning and natural language processing, significant advancements have been made in SLR, improving its accuracy and real-world applicability. Specifically, Continuous Sign Language Recognition (CSLR) is a challenging task as it aims to translate video captured gestures into meaningful text sequentially. Unlike word level sign recognition, CSLR should handle complex spatial-temporal dependencies, coarticulation effects, and variations in signer style. This paper provides a critical review of recent advances in CSLR, focusing on deep learning-based approaches, dataset challenges, feature extraction techniques, and evaluation metrics. We conclude the review by providing potential future research directions based on the limitations in the current methodologies.

Keywords: Sign language recognition, deep learning, computer vision, natural language processing, SLR datasets.

1 Introduction

There is an ever-increasing number of people with hearing loss across the world which highlights the need for a better communication medium between hearing and hearing impaired. According to world health organization [1], around 1.5 billion people will suffer with hearing related problems by 2021 and is projected to reach 2.5 billion by 2050. Accompanying individual hearing impaired with a dedicated human sign language interpreter for social interactions is not only difficult but also commercially impossible. Therefore, SLR becomes a crucial tool for bridging gap between normal and hearing-impaired people. SLR effectively address the translation of 1/2/3D signals into text. The 1D signals are obtained with hand gloves [2] fixed with motion detection sensors. These signals are pre-processed and classified using machine learning algorithms. The 2D signals are video sequences of the gestures which have attracted a huge research interest. This involves image segmentation, key point extraction, hand trajectory tracking, visual feature representations, and temporal mod-

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elling [3]. The skeletal based 3D representations of sign language capture the joint information of the hand movements for processing [4]. Though challenging, the 2D video based SLR is practised extensively across the research communities of computer vision and machine learning due to data availability and economical viability.

SLR models are divided into two categories based on the captured data. The first and less difficult is the word level sign language (WSLR) [5]. The second and hard one is continuous sign language recognition (CSLR) [6]. The core difficulties common in both models are sign to text conversion of hand gestures, movement trajectories, hand positions with respect to body and facial expressions. Word-level sign language recognition (WSLR) and continuous sign language recognition (CSLR) differ significantly in processing. In WSLR, isolated videos of labelled signs treat each of the following as an independent classification task. Whereas in CSLR the sign video is a sentence requiring additional processing with an alignment module [7]. The alignment module learns context aware temporal segmentation between the visual sequences and the gloss labels. Glosses are sequence of text giving a meaningful representation of the signs in the continuous video sequence. The datasets in WSLR are labelled clearly with each label representing an isolated video sequence. However, the CSLR datasets are continuous video sequences of signs representing a gloss label. Finally, the recognition of WSLR is a done with a classifier whereas in CSLR uses sequence modelling techniques for translating the recognized glosses into full sentences. This paper presents a critical analysis of datasets, methods, recognition techniques and evaluation metrics of latest CSLR models.

There are significant reviews on sign language recognition as a whole in the past years [8][9][10]. Most of these works lack the basic information on the latest models such as transformers, and other have completely excluded the importance of feature extraction in both visual and text modalities. Additionally, these works restrained from discussing the critical aspects of limitations of the latest models that could help the readers discover their future potential. Fig.1 shows the evolution of sign language representations from its inception to till date.

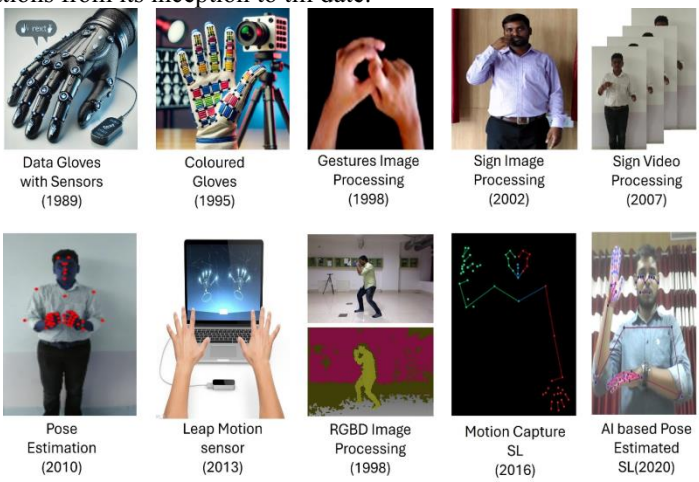


Fig. 1. Research progress and evolution of sign language representations over the years.

2 Continuous sign language recognition – Datasets and Evaluation Metrics.

2.1 Datasets

Large scale and well annotated video datasets are a leverage for SLR with deep learning models. The objective of a SL dataset is to train robust deep learning models for generalization across signers, environments and styles. This review examines some of the widely used SL datasets across both WLSLR and CSLR. Table 1 summarizes the video datasets only with their descriptive analysis. These datasets are most widely used by the researchers in the past decade. However, there are around 12 online SL datasets with word level representations which are not frequently used. Only 3 CSLR datasets are available online out of which RWTH-PHOENIX and Chinese SL are most widely used in research. Despite growth made in dataset development, challenges such as data sample limitations in continuous signing datasets, lack of signers diversity, limited multi modal annotations and real – world applicability. To advance CSLR research, future datasets should have larger and more diverse sign language corpora with open-source standardization and benchmarking.

Table 1. Video based sign language datasets in WLSLR and CSLR.

Dataset Name	Type	Language	Descriptions	Size	Annotations	Challenges
RWTH-PHOENIX-Weather 2014T	CSLR	German	Containing weather news broadcasts in real time.	8,257 video sequences, covering a vocabulary of 1,066 signs and 9 signers.	Includes gloss annotations, sign segmentations, and transcriptions.	Limited domain (weather news), relatively small vocabulary, and fixed background.
CSL-Daily	CSLR	Chinese	A large-scale dataset with fixed backgrounds.	21,000 videos, covering 1,000 daily vocabulary signs with 10 signers.	Includes sentence-level annotations and signer-independent splits.	Limited signer diversity and lack of contextual conversation-based signing.

MS-ASL	WLSL R	American	(Microsoft American Sign Language Dataset)	Over 25,000 samples from 222 signers, covering 1,000 unique signs.	Gloss-based annotations and signer metadata.	Limited continuous signing samples, mostly isolated signs.
WLASL (Word-Level American Sign Language)	WLSL R		A benchmark dataset for isolated ASL recognition.	21,083 videos, covering 2,000 words.	Word-level gloss annotations and signer-independent splits.	No continuous sentence signing and limited facial expression information.
LSA64	WLSL R	Argentinian	An Argentinian Sign Language dataset consisting of isolated sign videos.	3,200 video samples of 64 signs from 10 signers.	Gloss-based annotations.	Small dataset size and limited variation in signing styles.

2.2 Metrics

Word level Sign language videos consist of sequential frames, making convolutional neural networks (CNNs) like GoogleNet [11] and ResNet [12] popular for spatial feature extraction. Temporal modelling is achieved using networks such as recurrent neural networks (RNNs) [13] and temporal convolutional networks (TCNs) [14]. Some models, like C3D and I3D, use 3D CNNs to extract spatio-temporal features simultaneously. More recently, Transformer models and their variants have gained attention for sign language recognition, achieving improved performance. Metrics like accuracy, recall, precision and F1 score are used for evaluating the WLSLR models. The most widely used metric for evaluating CSLR is Word Error Rate (WER). It is defined as

$$wer = \frac{substitution+deletion+insertion}{Total\ words\ in\ Ground\ Truth\ sentences} \quad (1)$$

The wer measures the deviations in the predicted word sequence from the original ground truth sequence, quantifying the number of errors made by the trained model.

The substitution is the number of words needed to be substituted for incorrect predictions. Deletions are words that are missed by the model in the predicted sequence. Lastly, insertions are number of extra words added to the target sequence by the model. A lower WER indicates better recognition performance. WER in CSLR has limitations as it ignores sign language grammar, relies on gloss-level evaluation, and lacks context sensitivity, making it less suitable for real-world applications. Additionally, Bilingual Evaluation Understudy (BLEU) [15] evaluates the quality of machine-generated text. It measures how closely the generated text matches a set of reference texts by calculating the overlap of **n-grams** (sequences of words). BLEU is defined as

$$BLEU = Brevity\ Penalty \times exp\left(\sum_{n=1}^{N} w_n log(p_n)\right) \quad (2)$$

where, Brevity Penalty represents the length mismatches between generated and reference sentences. p_n is the n-gram precision calculated as the number of n-grams in the generated text appearing in the reference text. w_n are the weights assigned to different n-gram orders. It is particularly useful when CSLR models aim to generate natural language sentences instead of gloss sequences. Relies solely on n-gram matching, without deep semantic understanding. This metric is used scarcely by the CSLR researchers.

3 CSLR – Deep Learning based Recognition Methods

Deep learning-based recognition methods for WLSLR and CSLR captures a larger feature space, leading to better recognition accuracy when compared traditional machine learning based models. This section reviews the methods for CSLR using spatial, temporal and hybrid features. In addition to CNNs, RNNs, transformers, we present the latest trend of using multi modal fusion to leverage the characteristics of both visual and text features.

3.1 Methods based on CNN and RNNs

CNNs have proved to an excellent feature extraction tool on images which is leveraged to frames of videos in CSLR [16]. They capture good spatial information that is ingrained in the video frames. To extract temporal information that defines the changes in the hand and body movements of the signer, recurrent phenomena is engaged on the spatial features. This builds a combination of both CNN – RNN model for CSLR [17][18]. The combination has shown to extract spatio – temporal visual features representing the sign videos. Models like Resnet – 18 [19] and Resnet – 50 [20] were used as baseline models in CSLR. Similarly, RNN models that follow the CNN are long short-term memory (LSTM) [21] and gated recurrent units (GRU) [22]. Table 2 shows the latest CNN – RNN models used in building CSLR. Subsequently, pre-trained models like GoogleNet [11] and hybrid approaches, such as CNN-LSTM-HMM [23], improved performance but increase complexity. Studies explore different CNN-HMM structures for computational efficiency. Some approaches use motion

images [24], skeleton data [25], and 3D-CNNs [26] for enhanced feature extraction. Recent methods incorporate self-emphasizing modules to enhance spatial and temporal feature learning [27], improving sign recognition accuracy.

Table 2. CSLR methods based on CNNs, RNNs, HMMs, and their combination on RGB datasets.

Paper Title/ reference	Dataset	Methods	Performance reported (wer)	Merits	De-merits
Weakly Supervised Learning with Multi-Stream CNN-LSTM-HMMs	RWTH-PHOENIX-Weather 2014 T	CNN + LSTM + HMM	26.0%	Effectively learns from weak labels; integrates multiple streams (mouth and hand shapes) for better accuracy.	Requires complex multi-stream synchronization; training can be computationally intensive.
Self-sufficient Framework for Continuous Sign Language Recognition	RWTH-PHOENIX-Weather 2014 T	CNN + LSTM	22.3%	Does not require extra annotations; achieves state-of-the-art performance among RGB-based approaches.	May not fully capture non-manual features like facial expressions.
Sign Language Recognition Using Deep CNN	RGB Images - Custom Dataset	Deep CNN	24.31%	Employs deep CNNs for feature extraction, facilitating accurate recognition of static signs.	Limited to isolated sign recognition; does not address temporal dynamics in continuous signing.
Continuous Sign Language Recognition Algorithm Based on Object Detection	RGB Video	CNN + RNN	23.5%	Utilizes object detection to focus on relevant regions, improving recognition accuracy.	Performance metrics like WER not provided; applicability to large-scale datasets untested.
Recurrent	RWTH-	CNN +	20.22%	Effective for	training and

convolutional neural networks for continuous sign language recognition by staged optimization	PHOENIX-Weather 2014 T	TCN + Bi-LSTM + CTC		continuous sign language recognition without requiring frame-level annotations.	inference slower and more resource-intensive
SF-Net: Structured feature network for continuous sign language recognitio	CSL	CNN + Bi-LSTM	22.8%	Captures bidirectional temporal dependencies	Less effective for modelling very complex temporal dependencies
Fully convolutional networks for continuous sign language recognition	RWTH and CSL	2D CNN + 3D CNN + LSTM + CTC	21.89%	Spatial, spatio temporal and sequential dependencies are modelled effectively	requires high-performance hardware
	CSL	CNN + Encoder - Decoder	22.3%	Improved Sequence Generation	Encoder-decoder approach may struggle with variable-length sequences
	RWTH	C3D + Encoder - decoder	21.42%	Rich Spatio-temporal Feature Extraction	High Computational Demand: C3D models are computationally expensive

3.2 Methods based on Transformers

In recent years, Transformer models and their variants have gained immense popularity in feature extraction for CSLR and continuous sign language translation (CSLT) [28]. Originally developed for natural language processing (NLP) [29], transformers excel at capturing global dependencies within large datasets. Given sign language's unique grammatical structure, researchers have extended Transformer-based architectures to CSLR and CSLT tasks, often integrating both into a single framework. CSLR generates gloss sequences that follow the same order as sign language gestures

whereas CSLT converts gloss sequences into spoken language text, requiring a reordering of the sequence to align with natural language syntax.

In Multi-Branch Network (Shin et al.) [30], a combination of convolutional layers and transformer layers were employed to extract both local and long-range dependencies in sign language sequences. Video Transformer Network (VTN) (De Coster et al.) [31] trained a vision transformer (ViT) using features extracted from ResNet-34 (pretrained on ImageNet) and pose data obtained from OpenPose BODY135. Evaluated on AUTSL isolated sign recognition dataset, achieving 92.92% accuracy. The study uniformly down sampled video frames, which could lead to loss of important sign features, as sign language gestures are not evenly distributed in time. Another noticeable model was Mask-Future Transformer (Du et al.) [32], which applied swin-transformer for semantic features and proposes a Mask-Future Transformer for temporal modelling by computing self-attention without future frames to simulate real-time comprehension. Despite better performance, the two transformers made training difficult and computationally expensive. Transformer models have revolutionized sign language recognition and translation by efficiently capturing spatial and temporal dependencies. Hybrid models combining CNNs with Transformers offer state-of-the-art performance, but challenges like computational cost and data processing still need to be addressed. Future research should focus on efficient Transformer architectures and better real-time applications for sign language understanding.

3.3 Methods based on Multi Modals

Spatio-temporal visual features are crucial for Continuous Sign Language Recognition (CSLR) as they capture both spatial and temporal context in sign language videos. Recent advancements in feature extraction models have focused on improving context representation across frames to enhance recognition accuracy. This summary explores key methodologies, challenges, and improvements in spatial-temporal modelling, motion feature extraction, and cross-modal learning for CSLR. Temporal Difference Networks (TDN) [33] for Visual Features was applied to CSLR video data to learn the difference between consecutive frames and segments. The computed differences between frames introduce deviation in weight computation, affecting the overall quality of extracted visual features. Temporal Semantic Pyramid (TSM) [34] network addresses the semantic inconsistencies between frames using inter-scale and intra-scale attention mechanisms with increased computational costs.

To overcome the above shortcoming, multi modal methods were proposed to align gloss features with spatial feature extraction by computing contrastive cross-modal loss. Contrastive cross-modal loss [35][36] balances redundant information between visual features and BERT-extracted gloss features. This loss optimizes the extracted visual features, making them more robust for CSLR [37]. Inspiration from Multi-Modal Learning in Vision and Language was used in CSLR to construct graph embeddings for improving representation learning [38]. Recent works in image captioning and image-text matching were used to optimize multi-modal loss between visual and textual embeddings in CSLR [39][40]. Pre-trained language models like BERT and T5 have Transformer-based architectures that efficiently capture contextual de-

pendencies in text making them suitable for gloss feature extraction. The Masked Language Model (MLM) approach is used to fine-tune BERT [41] on gloss sequences, extracting rich gloss embeddings as labels for the visual feature extraction model [42].

The recent and latest works highlights the evolution of spatio-temporal and motion feature extraction techniques in CSLR and proposes an advanced gloss-aligned feature extraction method with contrastive cross-modal loss. By leveraging pre-trained language models like BERT for gloss feature learning, this approach enhances visual-text alignment and improves CSLR performance. Future work could explore further optimizations in multi-modal learning and efficient training strategies to reduce computational costs.

4 Recognition in CSLR

The isolated SLR is treated as a classification task, typically solved using SoftMax or SVM. In contrast, CSLR involves recognizing a sequence of signs, making it a sequence-to-sequence (seq2seq) problem. Two primary approaches adapted from natural language processing (NLP) for continuous SLR are Connectionist Temporal Classification (CTC) and the Encoder-Decoder Network. These methods help effectively model and recognize sequential sign language gestures.

4.1 Connectionist temporal Classification

Connectionist Temporal Classification (CTC) was first introduced by Graves et al. (2006) [43] as a solution for sequence-to-sequence (seq2seq) problems without requiring fine-grained labels. It became widely used in weakly supervised learning, particularly for tasks where input and output sequences have different lengths, such as speech recognition and continuous sign language recognition (CSLR).

CTC introduces an extra unit called 'blank' in the SoftMax output layer in addition to the dictionary L, which contains all words/signs in the dataset. The new dictionary is $L' = L \cup \{blank\}$, allowing the model to map sequences of different lengths without the need for frame-wise annotations. The probability of observing a particular output sequence π given an input sequence x of length π is computed as:

$$P(x) = \prod_{t=1}^{T} y_{\pi_t} \quad (3)$$

Further, it uses many-to-one mapping via the ‘Blank’ Token that removes blanks and repeated labels to generate possible output sequences using the B function. For example, B(-a-ab-) = B(-a-abb) = "aab". Here, "-" represents the blank token, which helps in handling different possible alignments of the same word. The probability P(l|x) of a particular labelling l given an input sequence x is computed as the sum of probabilities of all valid output sequences that map to l:

$$P\left(\frac{l}{x}\right) = \sum_{\pi=B^{-1}(l)}^{\square} P(x) \quad (4)$$

CTC aims to maximize the conditional probability of the target label sequence by using a forward-backward algorithm, which is based on dynamic programming. This enables efficient computation of probabilities across all valid alignments. Several research papers ([44], [45], [46], [47], [48], [49], [50]) have used CTC as the objective function for training CSLR models. CTC is widely used in weakly supervised learning because it does not require precise frame-level annotations.

However, the CTC assumes that each frame in the target sequence is conditionally independent, meaning the model does not explicitly account for dependencies between consecutive frames, which is unrealistic in CSLR, as frames within a local region of a sign language video are intrinsically linked due to the continuity of gestures. Due to this limitation of CTC, researchers began developing Encoder-Decoder Network architectures, which provide a more flexible seq2seq mapping. They allow the model to explicitly learn dependencies between frames, overcoming the independence assumption of CTC. Table 3 summarizes recent research on sign language recognition. Accuracy varies based on dataset complexity and model architecture.

Table 3. Comparative Analysis of Sign Language Recognition Models

Publication	Year	Methodology (Features & Classifier)	Dataset	Accuracy
Miah, A. S. M. [51]	2024	Graph Convolutional with Attention and Residual Connection	Word-Level American Sign Language (WLASL) Dataset	90.31%
			PAKISTANI SIGN LANGUAGE DATASET (PSL)	94.10%
			Mexican Sign Language (MSL) dataset	99.75%
			American Sign Language Lexicon Video (ASLLV) Dataset	34.41%
Xiao, Q., Li [52]	2024	standard Variational Autoencoder (VAE) + Generative Adversarial Networks (GAN) model	500 different isolated CSL words such as “happy” and “environment"	91.57% for 50 joints
			2000 SL words performed by over 100 actors	53.06% for 50 joints
			300 000 videos retrieved from YouTube. There are 400 human action classes, and each video in the dataset lasts about 10s	45.89% for 50 joints

Hax, D. R. T. [53]	2024	hybrid architecture consisting of a recurrent neural network (RNN), including a long short-term memory layer, on top of a convolutional neural network	6 different hand gestures with a total of 662 sequences	83.66%
Miah, A. S. M. [54]	2024	Graph meets with attention and CNN (GmTC)	Comprised ten distinct gestures from 14 individuals, with ten instances of each gesture, totaling 1400 unique data samples.	99.46%
			Consisted of 20 ASL words with a total of 18000 frames.	99.60%
			Encompassed 41 Japanese sign characters with 7,380 images adjusted to 400 x 400, captured from 18 individuals	92.37%
			Collected from 20 individuals, including 1,229 videos with 112,564 frames	99.30%
			Comprised 20 videos with 4-second recordings and two repetitions for each sign from each signer.	100%
			Consisted of 3200 videos with 10 non-expert subjects, each performing 5 repetitions of 64 unique sign types	99.10%

4.2 Encoder – Decoder Network

The Encoder-Decoder Network is widely used in CSLR to align input video sequences with target language sequences [55]. The encoder encodes the input into a fixed-size latent vector, while the decoder aligns and generates the output sequence. Early approaches relied on RNN-based architectures [56], but they suffered from long-term dependency issues, which led to the adoption of attention mechanisms [57] and Transformer-based models [58]. Transformers, being an Encoder-Decoder structure, use multi-head attention to globally link features, achieving strong performance in CSLR. Despite their effectiveness, Encoder-Decoder networks require high computational power and can be difficult to train. Techniques like pre-training, key pooling, and latent space optimization help improve convergence. Additionally, most encoder-decoder networks lack bi-directional decoding, which may cause information loss.

While BERT can address this, it comes at a high computational cost, making it impractical for many researchers. A trade-off between efficiency and accuracy is necessary for real-world applications.

5 Conclusions

Continuous Sign language recognition (CSLR) is a essential research area with significant social impact, as it facilitates communication between the hearing impaired and normal people. Automated SLR plays a key role in bridging this gap, making advancements in this field highly relevant. This review provides an overview of sign language recognition, covering traditional methods and focusing more elaborately deep learning-based approaches for CSLR. It examines key aspects such as datasets, evaluation metrics, methods, challenges, and future research directions. We highlight the latest state-of-the-art methods, particularly transformer-based and multi modal methods. Additionally, we compare the advantages and limitations of various recognition techniques. We conclude that the increasing availability of large datasets will enable more reliable validation of deep learning models, with significant challenges such as signer independence, model complexity, and real-world applicability. The in-depth analysis and proposed potential research directions described in this review can hopefully help in addressing these challenges.

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